

A counterexample to Frobenius-normalized cut-distance spectral continuity

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Source and question

The source paper is Vladimir Nikiforov, *Cut-norms and spectra of matrices*, arXiv:0912.0336. Theorem 3 of the source proves that, for entrywise bounded matrices $A \in \mathcal{M}_{m,n}$ and $B \in \mathcal{M}_{p,q}$,

$$\max_i \left| \frac{\sigma_i(A)}{\sqrt{mn}} - \frac{\sigma_i(B)}{\sqrt{pq}} \right| \leq 6 \delta_{\Xi}(A, B)^{1/2}.$$

In the concluding remarks, item 2 observes that the vector of singular values becomes a unit vector after division by the Frobenius norm and asks whether Theorem 3 can be modified accordingly.

Concluding remarks

1. Note that the norm $\|A\|_{\square}$ seems subtler than $\|A\|_{\square}$. Yet while $\|A\|_{\square}$ was used in a successful version of Szemerédi's Regularity Lemma ([11]), $\|A\|_{\square}$ has never been studied explicitly in this respect. A natural question arises: what type of Regularity Lemma one can prove using $\|A\|_{\square}$.
2. Let $A \in \mathcal{M}_{m,n}$ and $B \in \mathcal{M}_{p,q}$. We studied the spectral difference of A and B in the form

$$\max_{1 \leq i \leq \min(p,q,m,n)} \left| \frac{\sigma_i(A)}{\sqrt{mn}} - \frac{\sigma_i(B)}{\sqrt{pq}} \right|$$

On the other hand, the vector of the singular values of a matrix A becomes a unit vector when divided by $\|A\|_F$. Thus, it seems more appropriate to study

$$\max_{1 \leq i \leq \min(p,q,m,n)} \left| \frac{\sigma_i(A)}{\|A\|_F} - \frac{\sigma_i(B)}{\|B\|_F} \right|.$$

Can Theorem 7 be modified accordingly? Similar modifications seem possible for the eigenvalues of Hermitian matrices.

3. Some, but not all, of our results can be extended for complex graphons, i.e., measurable functions $f : [0,1]^2 \rightarrow \mathbb{C}$. One way of doing this is approximation by step functions with finitely many steps. We leave these extensions to interested readers.

References

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Result

There is no analogue of Theorem 3 in which the right-hand side is only a modulus of the cut-distance and the singular values are normalized by Frobenius norm. This failure occurs even for square 0-1 matrices of the same size and the same Frobenius norm.

Proof intuition

The cut norm is normalized by the total number of entries, so it sees a single dense row as negligible: a row with n ones in an $n \times n$ matrix has cut norm only $1/n$. The Frobenius-normalized spectrum sees the same row as a rank-one matrix whose entire Frobenius mass is in its first singular value. Comparing this star row with the identity matrix gives two matrices that are cut-close to zero, and hence cut-close to each other, but whose normalized singular spectra have completely different shape.

Theorem 1. *Let $\omega : [0, \infty) \rightarrow [0, \infty)$ be any function with $\omega(t) \rightarrow 0$ as $t \downarrow 0$. For all sufficiently large n , there are $n \times n$ 0-1 matrices A_n, B_n , with $\|A_n\|_F = \|B_n\|_F$, such that*

$$\delta_{\square}(A_n, B_n) \leq \frac{2}{n}$$

but

$$\max_{1 \leq i \leq n} \left| \frac{\sigma_i(A_n)}{\|A_n\|_F} - \frac{\sigma_i(B_n)}{\|B_n\|_F} \right| \geq 1 - \frac{1}{\sqrt{n}}.$$

Consequently, Frobenius-normalized singular spectra are not continuous with respect to the cut-distance alone.

Proof. Let

$$A_n = e_1 \mathbf{1}^T, \quad B_n = I_n,$$

where $e_1 = (1, 0, \dots, 0)^T$ and $\mathbf{1} = (1, \dots, 1)^T$. Both matrices have exactly n nonzero entries, all equal to 1, hence

$$\|A_n\|_F = \|B_n\|_F = \sqrt{n}.$$

We first bound the cut-distance. Since the matrices have the same size,

$$\delta_{\square}(A_n, B_n) \leq \widehat{\delta}_{\square}(A_n, B_n) \leq \|A_n - B_n\|_{\square}.$$

For the star row A_n , every submatrix sum is at most n in absolute value, so $\|A_n\|_{\square} = 1/n$. For $B_n = I_n$, a cut can contain at most n diagonal ones, so $\|B_n\|_{\square} = 1/n$. Therefore

$$\|A_n - B_n\|_{\square} \leq \|A_n\|_{\square} + \|B_n\|_{\square} = \frac{2}{n}.$$

It remains to compare the Frobenius-normalized singular spectra. The matrix $A_n = e_1 \mathbf{1}^T$ has rank one and its only nonzero singular value is

$$\|e_1\|_2 \|\mathbf{1}\|_2 = \sqrt{n}.$$

Thus

$$\frac{\sigma_1(A_n)}{\|A_n\|_F} = 1.$$

The identity matrix B_n has n singular values all equal to 1, and hence

$$\frac{\sigma_1(B_n)}{\|B_n\|_F} = \frac{1}{\sqrt{n}}.$$

The displayed lower bound follows from the first singular value alone. □

Verification notes

The proof uses only exact singular values and the elementary cut-norm estimates above. A small script in `code/verify_counterexample.py` checks the formulas and brute-forces the cut norm for small n .

Limitations and novelty check

This disproves the cut-distance-only Frobenius-normalized analogue of Theorem 3. It does not rule out variants with extra hypotheses, for example lower bounds tying $\|A\|_F$ and $\|B\|_F$ to $\sqrt{mn}$ and $\sqrt{pq}$, or density/pseudorandomness assumptions excluding matrices supported on $o(n^2)$ entries.

The bounded novelty check searched the run indexes for arXiv:0912.0336 and phrases around cut norms, spectra, Frobenius normalization, and singular values. Web searches for the exact paper title with “Frobenius norm singular values”, for “Frobenius cut-distance singular values Nikiforov”, and for the exact concluding-question phrase found the source arXiv page but no later public answer to this specific obstruction.

References

V. Nikiforov, *Cut-norms and spectra of matrices*, arXiv:0912.0336.